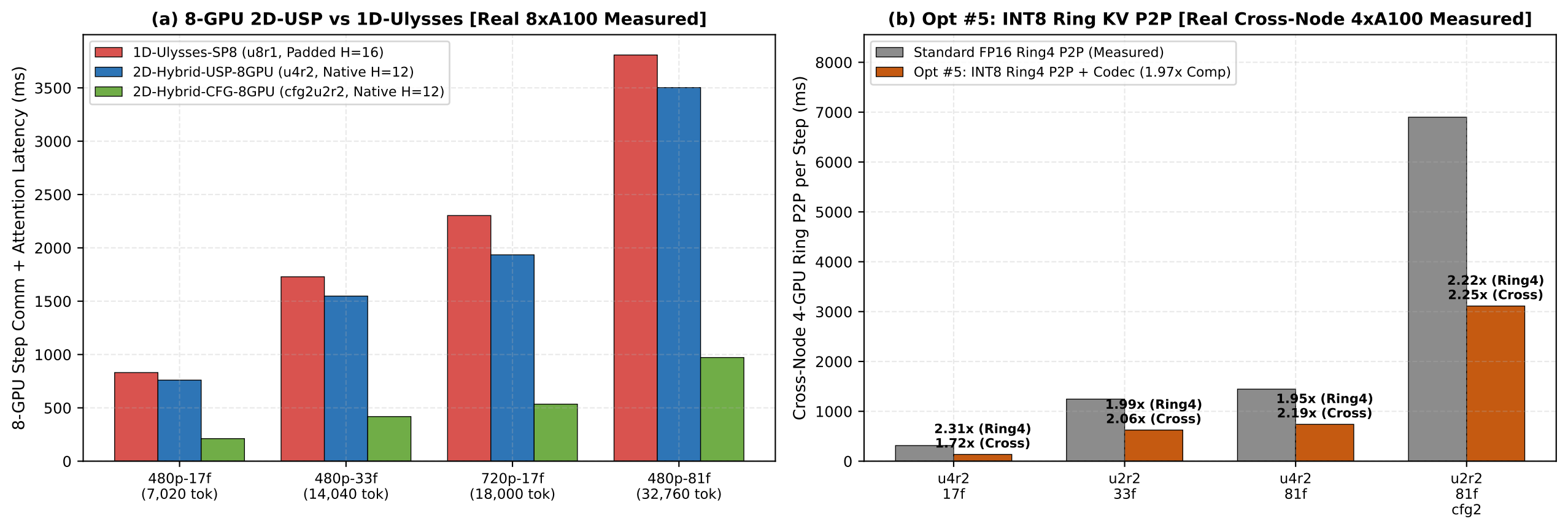
(a) 8-GPU 2D-USP vs 1D-Ulysses [Real 8xA100 Measured]
1D-Ulysses-SP8 (u8r1, Padded H=16)
2D-Hybrid-USP-8GPU (u4r2, Native H=12)
2D-Hybrid-CFG-8GPU (cfg2u2r2, Native H=12)
8-GPU Step Comm + Attention Latency (ms)
0
500
1000
1500
2000
2500
3000
3500
480p-17f
(7,020 tok)
480p-33f
(14,040 tok)
720p-17f
(18,000 tok)
480p-81f
(32,760 tok)
(b) Opt #5: INT8 Ring KV P2P [Real Cross-Node 4xA100 Measured]
Standard FP16 Ring4 P2P (Measured)
Opt #5: INT8 Ring4 P2P + Codec (1.97x Comp)
Cross-Node 4-GPU Ring P2P per Step (ms)
0
1000
2000
3000
4000
5000
6000
7000
8000
2.31x (Ring4)
1.72x (Cross)
1.99x (Ring4)
2.06x (Cross)
1.95x (Ring4)
2.19x (Cross)
2.22x (Ring4)
2.25x (Cross)
u4r2
17f
u2r2
33f
u4r2
81f
u2r2
81f
cfg2